I have a model that applies alternating magnetic fields to a conducting sphere. It calculates the induced currents from fields normal to the surface. It relates currents to emitted radial fields, which in turn affect the current. It solve this self-consistently to get currents and net fields that locally have currents given by the rate of change of the normal fields including the emitted fields.

Does this approach implicitly include the inductive affects of emitted on fields on the admittance that the current sees on the surface, or does the magnetic inductance need to be added in to the model?

Note that the fields are only calculated at the surface.

Thought for ...

If you're solving **self-consistently** for the surface currents where

1. the *driving* quantity is the **time derivative of the normal magnetic field at the surface**, and
2. that "normal field" is explicitly the **total** normal field = applied + field produced by those same currents,

...then you are *already building in* the inductive back-reaction, in the sense that you're including the **magnetic field generated by the currents** and letting it modify the thing that drives the currents.

But there's a big "however," and it hinges on *what you mean by "currents from fields normal to the surface"* and what operator you're using.

The clean EM way to think about it

For eddy currents in a conductor, the fundamental driver is **Faraday's law**:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Currents respond to the **tangential electric field** at the surface (or just inside it), through a surface impedance/admittance relation (skin depth physics):

$$\mathbf{J}_s = Y_s\,\mathbf{E}_t \quad \text{(or)} \quad \mathbf{E}_t = Z_s \mathbf{J}_s.$$

The *inductive* part of what the current "sees" is basically the $-\partial\mathbf{A}/\partial t$ piece of $\mathbf{E}$ (plus gauge/charge terms), and $\mathbf{A}$ is generated nonlocally by the currents. So inductance is not a bolt-on element in Maxwell; it's what you get when you correctly include the field produced by currents in Faraday's law / the vector potential.

So:

- **If your self-consistency uses the full EM mapping** "surface current distribution → magnetic field everywhere on the surface (and thus the induced EMF)" then *that is the inductance*.
- You generally **do not add a separate inductance** on top of that; doing so would usually *double count*.

The part that can be missing when you only use B_n on the surface

Inductance is fundamentally **nonlocal**: a patch of current affects the EMF seen by other patches through the vector potential / mutual inductance kernel.

So the key question becomes:

Are you using a nonlocal operator from current → emitted field?

- **Good (includes inductive coupling):** you compute the emitted field from the entire surface current distribution via something Biot–Savart / vector-potential / boundary-integral-ish (even if you only *evaluate* it on the surface). That builds a dense coupling between modes/patches. In that case, the inductive effects are implicitly there.
- **Risky (can miss inductance):** you use a *local* rule like "current at a point is proportional to $\partial B_n/\partial t$ at that point" without a proper global coupling kernel. That tends to miss mutual inductance and can act like you've kept resistive/skin-depth physics but thrown away the "coil-ness."

Is B_n alone enough?

On a sphere, **it *can* be**, but only if you're doing it in a mode-complete way.

- If you decompose $B_n(\theta, \phi)$ into spherical harmonics, then for regions without currents (outside the sphere) the magnetic field is potential-like, and **the full external field is determined by the radial component at the boundary** (up to the usual interior/exterior coefficients). In that *special geometry*, working purely with B_r modes can still capture the correct inductive response—**provided** your mapping from currents → B_r is the one implied by Maxwell, and your current basis enforces $\nabla_s \cdot \mathbf{J}_s = 0$ appropriately.
- If you're not in a spherical-harmonic/mode framework and you only keep "radial field at the surface" but don't correctly reconstruct the tangential $\mathbf{E}_t$ / vector potential

effect, then it's easy to accidentally drop the inductive (reactive) part or get it wrong.

A quick tell: **does your model produce a reactive (imaginary) component** of the effective response that behaves sensibly with frequency and skin depth? If it's purely resistive-looking when it shouldn't be, inductive coupling is being lost somewhere.

So do you "need to add inductance"?

Usually: no, *if* your self-consistent field calculation is Maxwell-consistent.

If you're computing the emitted fields from the currents with the correct nonlocal kernel and feeding that back into the driving quantity (through Faraday / $\partial B/\partial t$), then the inductive effects are already embedded.

But: maybe, if your "admittance" is only the material skin-depth piece and your feedback path omits the nonlocal EMF.

A common split is:

- **Surface impedance** Z_s from skin depth: local, material property (gives resistance + a local reactive term).
- **Geometric inductance** from magnetic energy in space: nonlocal, comes from the field solution.

If your code's "admittance" is basically just the skin-depth boundary condition and your field-feedback is incomplete/localized, then you may be missing the geometric inductance part.

Two concrete sanity checks (high value, low pain)

1. **Compare against the analytic solution for a conducting sphere in a uniform oscillating field** (classic eddy-current problem).
   The induced dipole moment / shielding factor as a function of δ/R (skin depth over radius) is known and has a characteristic complex frequency dependence. If you match that, you're almost certainly not missing inductance.
2. **Check passivity / energy behavior:**
   For a passive conductor, the operator you build should yield:

- nonnegative dissipation (real power ≥ 0),
- and a reactive part consistent with stored magnetic energy (sign conventions aside).

Numerically, if your coupling matrix is wildly non-symmetric in a basis where it should be symmetric (or if the reactive part looks unphysical), that's a red flag that the inductive coupling kernel isn't represented correctly.

Bottom line

- If your "self-consistent" loop truly computes **the field produced by the currents** (even evaluated only on the surface) using a Maxwell-valid nonlocal operator, and you drive currents from the **total** $\partial B_n/\partial t$, then **you are already including inductive back-reaction**—no extra inductance term should be added.
- If your method uses **only local relationships** between B_n and current (or otherwise doesn't reconstruct the tangential EMF / mutual coupling), then you can be **missing geometric inductance**, and "adding inductance" is really a patch for a missing nonlocal coupling term.

If you tell me one detail—*are you representing currents in spherical harmonics (toroidal modes) or as surface patches, and how do you compute the emitted B_n from J?*—I can say much more definitively which side you're on and what to fix (if anything).

Yes I'm doing all the calculations in spherical field harmonics. Note the conductor is a thin surface of 0 thickens. No skin depth

Thought for ...

Got it — **spherical harmonics** + a **zero-thickness conducting surface** (a 2D sheet) changes the story in a nice way.

What "inductance" can even mean for a 0-thickness shell

With no skin depth / no volume conductor, you've basically removed:

- **internal inductance** (magnetic energy stored *inside* the conductor volume),
- and the classic diffusion/skin-depth physics (which normally gives a frequency-dependent complex surface impedance).

What's left is purely:

- **geometric inductance** = magnetic energy stored in the fields in the regions *inside and outside the sphere* that are generated by the surface currents.

So if anything is "missing," it would only be **that geometric inductive coupling** between the sheet current and the EMF it induces.

If you include the emitted field self-consistently, you are already including geometric inductance

In a spherical-harmonic formulation, each (ℓ, m) mode basically behaves like its own little "circuit":

- the surface current mode produces a magnetic field mode,
- that field implies an induced tangential E_t through Faraday's law,
- and the sheet obeys $\mathbf{K} = \sigma_s \mathbf{E}_t$ (or $\mathbf{E}_t = R_s \mathbf{K}$ with sheet resistance $R_s = 1/\sigma_s$).

If your self-consistent solve is truly:

> **total field at the surface** = applied + field from $\mathbf{K}$
> and the induced driving term is computed from the **time derivative of that total field**,
> and then $\mathbf{K}$ is solved such that it's consistent with the boundary condition / Ohm's law on the sheet,

...then you've already built in the inductive back-reaction. There is no "extra inductance" you should tack on — that would normally double-count.

Why this is especially clean in harmonics: for a perfect sphere, the mapping

$$\mathbf{K}_{\ell m} \;\longrightarrow\; B_{r,\ell m}(R)$$

is *nonlocal in real space* but becomes **diagonal in** (ℓ, m). That diagonal operator *is* the mutual/geometric inductance kernel in disguise.

The one trap: using $B_r(R)$ alone can be fine — but only if you're using the right operator

You said you "calculate induced currents from fields normal to the surface."

That's okay **only if** you're not using a purely local rule like "$K \propto \partial B_r/\partial t$" at the same point.

In Maxwell terms, the sheet responds to **tangential electric field**, and the tangential E is related to $-\partial \mathbf{A}/\partial t$ (plus any $-\nabla\phi$ piece, which often vanishes for the divergence-free eddy-current modes). The normal B_r at the surface *can* determine the mode amplitudes on a sphere, but the proportionality involves the **geometry factors in the harmonic operator**.

So the practical check is:

- If your algorithm for each ℓ effectively produces a **complex modal impedance**

$$Z_\ell(\omega) = R_s + i\omega L_\ell$$

  (with L_ℓ coming from the field solve), then you're good.
- If your "admittance" is only $1/R_s$ and the model never produces an $i\omega$ reactive term from the field coupling, then you've accidentally dropped geometric inductance somewhere.

For a 0-thickness sheet, **that** $i\omega L_\ell$ should be the *only* reactive part (aside from any displacement-current/capacitive physics you're explicitly neglecting).

Bottom line

No, you do not need to add magnetic inductance separately *as long as* your harmonic self-consistent field solve uses the sheet-current → emitted-field relation implied by Maxwell (which, on a sphere, should give you a diagonal-in-ℓ, m coupling). In that setup, the inductive effect is *already inside the operator* you're iterating/inverting.

If you want, tell me exactly what your constitutive step looks like for each ℓ, m (e.g., do you solve for a stream function $S_{\ell m}$, do you use $K = \hat{r} \times \nabla S$, and what equation links $\partial B_r/\partial t$ to E_t?). With that, I can point to the exact place geometric L_ℓ lives in your formulation and what its scaling with ℓ should look like.

How would I compute Z from my model?

Cool — in your spherical-harmonic, zero-thickness-sheet world, the clean way to "read off" an impedance is to treat **each** (ℓ, m) as a one-port: a generalized EMF drives a generalized current (the sheet-current mode amplitude), and your model already gives you the transfer function between them.

Here are two practical ways to compute $Z_\ell(\omega)$ from the stuff you already have.

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1) Modal impedance from surface power and current (most robust)

This avoids arguing about what "voltage" is on a sphere.

For a time-harmonic mode $e^{i\omega t}$, define the complex power delivered to the sheet:

$$P_\ell \equiv \frac{1}{2}\int_S \mathbf{E}_t \cdot \mathbf{K}^* \, dA$$

where $\mathbf{K}$ is surface current density (A/m) and $\mathbf{E}_t$ is tangential electric field (V/m).

Now define a modal “current magnitude” as:

$$I_\ell^2 \equiv \int_S |\mathbf{K}|^2 \, dA$$

Then the **modal impedance** is:

$$Z_\ell(\omega) \equiv \frac{2P_\ell}{I_\ell^2}$$

Why this works:

- If the sheet obeys Ohm’s law $\mathbf{E}_t = R_s \mathbf{K}$ (with sheet resistance R_s in Ω/sq), then
  - $\Re(Z_\ell) = R_s$ (nice sanity check).
- Any extra reactive part $\Im(Z_\ell)$ is coming from the **inductive back-reaction** your self-consistent EM solve produced (i.e., geometric inductance).

How to implement with harmonics:

For a single normalized vector spherical harmonic basis function $\mathbf{X}_{\ell m}$ (divergence-free toroidal one),

$$\mathbf{K} = k_{\ell m} \, \mathbf{X}_{\ell m}$$

and similarly $\mathbf{E}_t = e_{\ell m} \, \mathbf{X}_{\ell m}$ for the same mode (everything diagonal).

Then the integrals collapse to constants:

$$P_\ell = \frac{1}{2} e_{\ell m} k_{\ell m}^* \int |\mathbf{X}_{\ell m}|^2 dA, \quad I_\ell^2 = |k_{\ell m}|^2 \int |\mathbf{X}_{\ell m}|^2 dA$$

so the normalization cancels and you get the simplest result:

$$\boxed{Z_\ell(\omega) = \frac{e_{\ell m}}{k_{\ell m}}}$$

So in practice, you just need the **complex amplitude ratio** e/k for that mode.

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2) Impedance as "generalized voltage / generalized current"

Pick a conventional "voltage" for the mode as the line integral of $\mathbf{E}$ around a representative loop of that current pattern.

For toroidal modes it's natural to use a loop at fixed θ (a latitude circle). Define:

$$V_\ell(\theta) \equiv \oint \mathbf{E}_t \cdot d\boldsymbol{\ell}$$

$$I_\ell(\theta) \equiv \oint \mathbf{K} \cdot d\boldsymbol{\ell}$$

Then $Z_\ell(\theta) = V_\ell(\theta)/I_\ell(\theta)$.
For a pure single harmonic mode, this ratio should be independent of θ (up to zeros where the mode vanishes), and it will match the e/k result above.

This is more "circuit-y," but it's fussier numerically because you're sampling a specific loop.

What do you use for E_t if your model is in terms of $B_r(R)$?

Since you're solving in harmonics, you can get E_t mode amplitudes from Faraday's law in harmonic form.

A convenient identity (outside sources, quasi-static, no displacement current) is that for the toroidal electric field associated with changing poloidal magnetic flux, the tangential E is tied to $\partial B_r/\partial t$ via the surface curl:

$$(\nabla_s \times \mathbf{E}_t) \cdot \hat{\mathbf{r}} = -\frac{\partial B_r(R)}{\partial t}$$

For a single (ℓ, m),

$$\mathbf{E}_t = e_{\ell m}\, \mathbf{X}_{\ell m}, \quad B_r(R) = b_{\ell m}\, Y_{\ell m}$$

and the operator relation becomes (up to sign conventions)

$$\boxed{e_{\ell m} = -\frac{R}{\ell(\ell+1)}\, \frac{\partial b_{\ell m}}{\partial t}}$$

For harmonic time dependence $e^{i\omega t}$,

$$\boxed{e_{\ell m} = -i\omega\, \frac{R}{\ell(\ell+1)}\, b_{\ell m}}$$

So if your model gives you $b_{\ell m}$ (total radial field at the surface) and you have $k_{\ell m}$ (current-mode amplitude), you immediately get:

$$\boxed{Z_\ell(\omega) = \frac{e_{\ell m}}{k_{\ell m}} = -i\omega\, \frac{R}{\ell(\ell+1)}\, \frac{b_{\ell m}}{k_{\ell m}}}$$

This is the “extract Z from what you already compute” formula.

Sanity checks:

- If your sheet has finite R_s and you enforce $\mathbf{E}_t = R_s\mathbf{K}$, then Z_ℓ should have $\Re(Z_\ell) = R_s$ (independent of ℓ). Any extra ℓ-dependence in the real part usually means your normalization of $k_{\ell m}$ vs $\mathbf{K}$ is off.
- $\Im(Z_\ell)$ should scale like ω at low frequency (inductive), and the effective $L_\ell \equiv \Im(Z_\ell)/\omega$ should be positive (with your sign convention).

Quick recipe (what to do in code)

For each ℓ, m at a chosen ω:

1. Drive with a known applied field mode amplitude $b_{\ell m}^{\mathrm{app}}$.
2. Solve your self-consistent system → get total $b_{\ell m}(R)$ and sheet current amplitude $k_{\ell m}$.
3. Compute $e_{\ell m} = -i\omega \frac{R}{\ell(\ell+1)}\, b_{\ell m}$.
4. Output $Z_\ell = e_{\ell m}/k_{\ell m}$.
5. Optional: report $R_\ell = \Re(Z_\ell)$, $L_\ell = \Im(Z_\ell)/\omega$.

If you tell me what basis you use for surface currents (stream function $S_{\ell m}$? direct $\mathbf{X}_{\ell m}$ coefficient? something else), I can translate "your stored coefficient" into $k_{\ell m}$ unambiguously so you don't get bitten by a normalization factor.

So my model relates dBdt normal to electric field on the surface. Electric field is related to current density by the resistivity alone. Self consistency make it add up when we include the nonlocal fields due to harmonics. the fact that I relate by resistivity alone does not prevent inductance to be accounted for?

Yep — **relating E_t to K with resistivity alone does *not* block inductance**, as long as your *self-consistency* is done the Maxwell way (which, in spherical harmonics, it usually is).

Here's the intuition in one breath:

- The sheet law $\mathbf{E}_t = R_s\mathbf{K}$ is just the **material constitutive relation** (ohmic dissipation in the sheet).
- The **inductance** is not another material term you add to that law. Inductance comes from **Faraday's law + the fact that your currents create magnetic flux** (i.e., the nonlocal field kernel).
- When you compute the **total** B_n at the surface (applied + emitted) and feed $\partial B_n/\partial t$ into the equation that generates E_t, you are already including the "$-L\,dI/dt$" back-EMF effect — just expressed in field language.

So the inductive behavior lives in the loop:

$$\mathbf{K} \xrightarrow{\text{Biot–Savart / harmonic kernel}} B_n^{\text{emit}} \xrightarrow{\partial/\partial t} \nabla_s \times \mathbf{E}_t \xrightarrow{\mathbf{E}_t = R_s\mathbf{K}} \mathbf{K}.$$

That loop is literally "mutual inductance" in operator form.

What you're *not* including (by design)

Because you have a 0-thickness sheet with no skin depth, you are not modeling:

- diffusion of fields into a volume conductor,
- internal inductance / kinetic inductance (unless you explicitly put it in),
- any capacitive/displacement-current stuff.

But **geometric inductance** (stored magnetic energy in space due to the sheet currents) *is* included, provided your emitted-field operator is the correct Maxwell one.

The one subtle gotcha

If you enforce $\mathbf{E}_t = R_s \mathbf{K}$ *pointwise* and then separately compute emitted fields, you're fine **only if** the E_t you're using is the one implied by Faraday's law for the *total* B_n.

You said you do exactly that: *"model relates dBdt normal to electric field on the surface... self consistency includes nonlocal fields due to harmonics."* ✅ That's the right structure.

A quick internal consistency check in your harmonic basis:

- If you drive one (ℓ, m) at frequency ω, the solution for the modal current amplitude should look like

  $$K_{\ell m} = \frac{(\text{drive})}{R_s + i\omega L_\ell}$$

  for some positive L_ℓ coming from the field kernel.
- In other words, even though your **constitutive law** is purely resistive, your **overall input–output** becomes complex because the field coupling brings in the $i\omega$ (via the time derivative in Faraday's law).

If you're seeing that kind of phase lag / reactive behavior in the mode response, you've got inductance "for free."

When would you need to *add* an inductive term?

Only if the sheet itself has additional physics beyond simple ohmic dissipation, e.g.:

- **kinetic inductance** (superconducting film or very thin normal film at high frequency),
- a phenomenological surface impedance $Z_s(\omega)$ you want to impose that already includes some reactive part not produced by geometry.

Then you'd replace R_s with a complex $Z_s(\omega)$ in $\mathbf{E}_t = Z_s\mathbf{K}$. But you still wouldn't add a *separate* geometric inductance on top of your self-consistent field solve.

If you want, paste the exact harmonic equation you're using for "$\partial B_n/\partial t \to E_t$" (even just the ℓ-dependent prefactor). I can sanity-check that it's the standard $\ell(\ell+1)$ operator and tell you what L_ℓ scaling you should expect.